Advanced Labs - Theme C: Fabrication and Electrical Characterization of MOS Capacitors

Ibraheem Farzat
202118029

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1 MOS Capacitors fabrication process

The fabrication of Metal-Oxide-Semiconductor capacitors is a multi-step process that involves creating a capacitor structure on top of a semiconductor substrate, typically silicon. The following will be a description of the procedure that we carried out over three weeks.

1.1 Experimental Setup and Preparations

Before starting the fabrication process we first needed to make sure that we were not bringing in any contamination into the clean room with us. We had to wear dust-free clothes, shoes, and cap. In addition, we had to go through an air shower before entering and we used plastic gloves to handle any tools or chemical bottles.

The experimental setup was done by the professor and TAs beforehand and was as follows:

1. Si wafer: a 2cm square p-type silicon wafer is prepared with a mirror-polished (100) orientation.
2. Clean the wafer: RCA cleaning is performed on the silicon wafer to remove contaminants from the wafer surface.

1.2 Expirement Procedure

The experiment was as follows:

1. Gate Oxidation: This is the first step in the fabrication, we use Teflon tweezers to align the wafers in the grooves of a quartz boat in a quartz tray. While wearing rubber gloves, we use a pull-out rod to insert the tray into the mouth of $1000°C$ dry oxygen furnace for 10-15 minutes and then we use a quartz cap to cover the oxidation furnace. After the designated time we pull it out using the pull-out rod. The silicon reacts with oxygen in the furnace and creates a layer of silicon oxide. The desired thickness should be between 20-40 nm.
2. Al deposition: We place the Al seed wire and the sample into the evaporation system chamber, which uses a tungsten filament to heat up the Aluminum until it evaporates. After confirming that everything is working fine we start the current until the bell jar becomes opaque enough and the deposition has reached the desired thickness. Aluminum is deposited on the surface of the wafer. We stop the current and take out the wafer after making sure that all of the equipment are turned off.
3. Patterning: In this step, we add a photoresist, a chemical that reacts to UV rays to harden up, onto the sample. We use a centrifuge to ensure that the whole wafer is coated evenly. After this, UV light is shined through a mask with the desired electrode pattern as a mold to the circuit inside the MOS.
4. Etching: In this stage, we use a mixture of phosphoric acid and nitric acid to remove the unreacted photoresist by raising it in the solution. We also cleaned the sample using distilled water to remove any residual contaminants.
5. Resist peeling: Now we eliminate any residual organic material by submerging the sample in acetone and methanol for 5 minutes each and then with distilled water again for another 5 minutes.
6. Backside Processing: We use sandpaper to remove the oxide film on the backside of the sample, and then we apply a silver paste to bond it to a stainless-steel plate.
7. Evaluating electrical characteristics: We analyze the electrical characteristics of the MOS capacitor like the current-voltage and capacitance-voltage.

2 Analysis and Results

The electrical characteristics of the samples fabricated are shown in Fig 1.

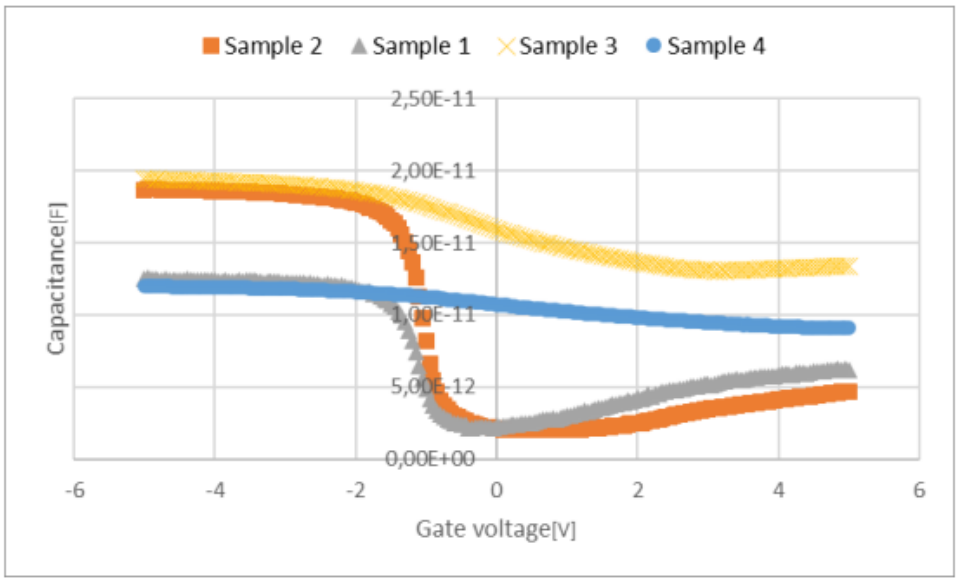


Figure 1: Electrical characteristics of the samples

From the figure, we can find the film thickness using the formula:

$$C_{ox}(F) = A\frac{\epsilon_{ox}}{t_{ox}}$$

Where:

- $C_{ox}(F)$ is the capacitance in the accumulation region.
- A is the gate electrode area.
- ϵ_{ox} is the SiO_2 dielectric constant.
- t_{ox} is the oxide film thickness.

Using the values from our data, we can find the oxide film thickness t_{ox}

$$t_{\text{ox}} = 10000 \times 10^{-6}\frac{3.45 \times 10^{-13}}{1.2 \times 10^{-11}} = 2.875 \times 10^{-4} \text{ m for samples 1 and 4}$$
$$t_{\text{ox}} = 10000 \times 10^{-6}\frac{3.45 \times 10^{-13}}{1.9 \times 10^{-11}} = 1.815 \times 10^{-4} \text{ m for samples 2 and 3}$$

It can be inferred that samples 1 and 4 are representative of the high impurity concentration samples, as evidenced by the pronounced gradient observed in the central region of the graph. Consequently, this implies that samples 2 and 3 are indicative of the low impurity concentration.

$$\frac{1}{C_D} = \frac{1}{C} - \frac{1}{C_{\text{ox}}}$$

$$4.76 \times 10^{11} - 1.21 \times 10^{11} = \frac{1}{C_D}$$

$$3.54 \times 10^{11} = \frac{W}{A\epsilon_{si}}$$

$$W = 3.72 \times 10^{-6}$$

$$W_{Dm} \approx \sqrt{\frac{4\epsilon_s kT \ln\left(\frac{N_A}{n_i}\right)}{q^2 N_A}}$$

$$3.72 \times 10^{-6} \approx \sqrt{\frac{4(1.05 \times 10^{-12})(1.38 \times 10^{-23})(295) \ln\left(\frac{N_A}{10^{10}}\right)}{(1.5 \times 10^{-19})^2 N_A}}$$

$$N_A \approx 1$$

3 Final thoughts

Over the past three weeks, my involvement in experiments on nanotechnology has been both fascinating and educational. I gained valuable insights into the complex processes involved in fabricating Metal-Oxide-Semiconductor (MOS) and MOS Field-Effect Transistors (MOSFET). These sessions highlighted the critical need for conducting such experiments in a clean, controlled environment. This is essential to avoid any contamination, like from dust particles, which could impact the results.

I also learned that creating these components involves careful handling of chemicals and a precise approach. Every step in the fabrication process is crucial and requires attention to detail.

Although my future career may not involve direct engagement in semiconductor manufacturing, participating in this experiment has greatly enhanced my appreciation for the intricate technology and sophisticated processes involved in creating such marvelous and complex devices. This experience has provided me with a deeper understanding and respect for the advanced techniques and precision required in the semiconductor industry, which play a crucial role in the development of cutting-edge technology that shapes our modern world.